

More sound. Less noise.

How can headphones make things quieter by creating another sound?



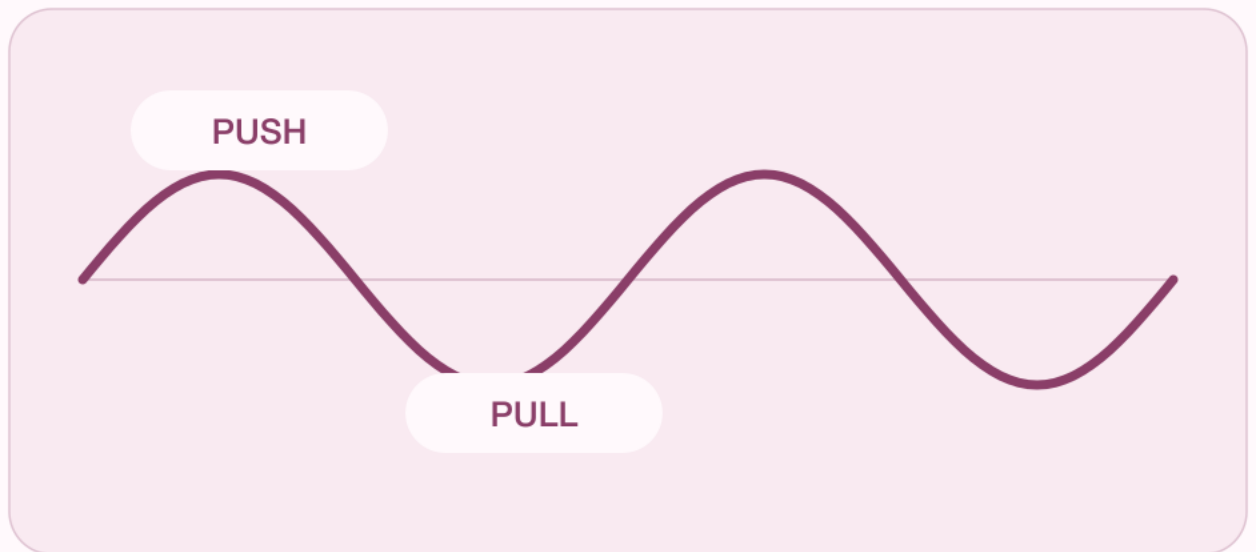
The wave makes it click.

START WITH SOUND ITSELF

Air pressure goes up and down.

Sound makes air pressure rise and fall.

A wave is a picture of that pressure changing.



The headphones cannot delete the air.

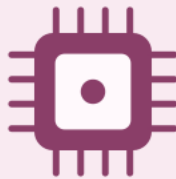
They can oppose part of its movement.

Measure noise. Make its opposite.

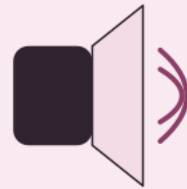
Microphones measure unwanted sound.
A processor works out a cancelling signal.



Microphone



Processor

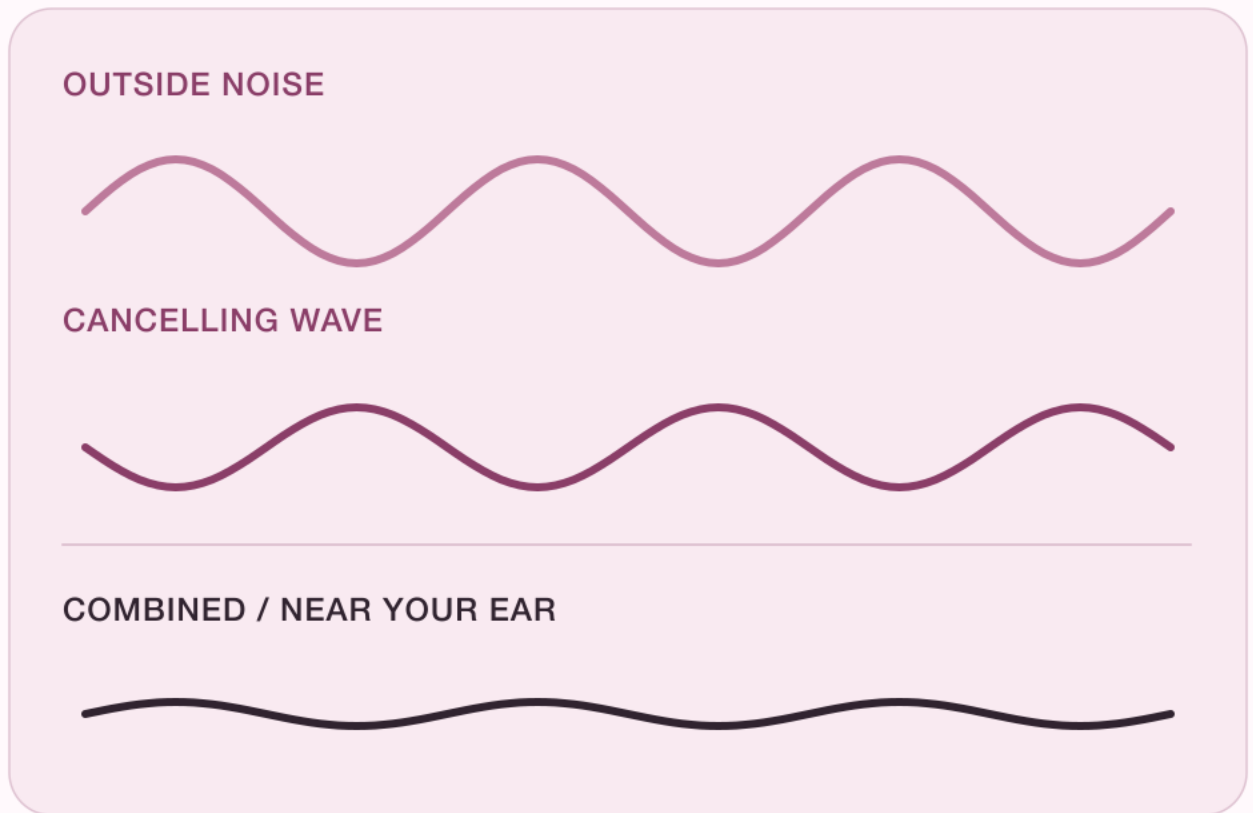


Speaker

The speaker creates an opposing sound wave.
That is active noise cancellation, or ANC.

THE SURPRISING PART

A push meets a pull. The result gets smaller.



Timing and strength have to match closely.

Real headphones reduce noise, not all sound.

Simplified wave diagram, not a measurement.

Your ears get quieter. The room does not.

The cancelling signal is designed for the place where you hear it.



The noise still exists outside the headphones.
ANC does not silence the whole plane or cafe.

Steady hum: easier. Sudden voice: harder.



STEADY HUM



CHANGING SOUND



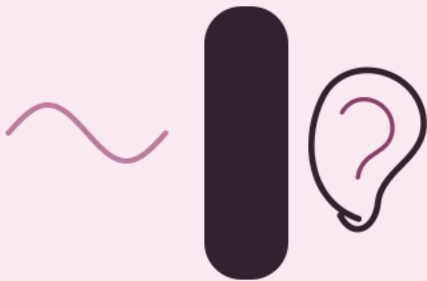
Low, steady noise is generally easier to reduce.

Voices and claps change fast and can remain.

The result depends on the headphones, fit and surroundings.

Cushions block. ANC opposes.

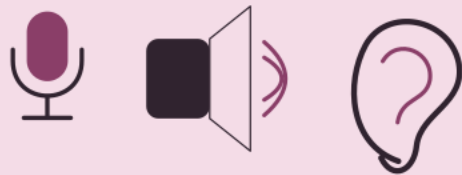
PASSIVE ISOLATION



A physical seal

No powered signal

ACTIVE CANCELLATION



An opposing wave

Electronics need power

Good cushions or ear tips help with the seal.
Active and passive reduction work together.

Music can be paused. ANC can stay on.



ANC

ON

MUSIC

PAUSED

Many ANC headphones work without music.
They still need battery power and a snug fit.

Use an awareness mode when you need to hear around you.

Listen. Oppose. Reduce.

01 Microphones measure the noise.

02 Speakers add an opposing wave.

03 Less pressure change reaches your ear.

Expect less hum.
Not guaranteed silence.